Supplementary Materials:

Domain-Specific Gender Gaps in Sexual Attitudes

[Blinded for Review]

S1. Variable Coding and Measurement

S1.1 Dependent Variables

Sexual attitude items are drawn from the CGSS morality module, which asks respondents to evaluate specific sexual behaviors on a scale from 1 ("always wrong") to 5 ("not wrong at all"):

- **Premarital sex**: "What is your opinion about sexual relations before marriage?" (a38)
- **Extramarital sex**: "What is your opinion about a married person having sexual relations with someone other than his or her spouse?" (a39)
- **Same-sex relations**: "What is your opinion about sexual relations between two adults of the same sex?" (a40)

Higher values indicate more accepting attitudes.

S1.2 Gender Ideology Index

The gender ideology index is constructed from five items (a421–a425), each rated on a 1–5 agree-disagree scale:

1. "Men should focus on career, women should focus on family" (a421)
2. "Men are inherently more capable than women" (a422)

3. "For women, marrying well is more important than doing well at work" (a423)
4. "When jobs are scarce, women should be fired first" (a424)
5. "Housework is primarily women's responsibility" (a425)

The index is the mean of all available items (Cronbach's $\alpha = 0.72$). Higher values indicate more patriarchal gender ideology. For regression models, the index is standardized (z-scored).

S1.3 Control Variables

- **Age**: Continuous
- **Age^2/100**: Quadratic age term divided by 100 for scaling
- **Education**: Years of formal schooling (continuous)
- **Married**: Binary (1 = currently married or cohabiting)
- **CCP membership**: Binary (1 = member of the Chinese Communist Party)
- **Religious**: Binary (1 = identifies with any religion)
- **Internet user**: Binary (1 = uses the Internet)
- **Urban**: Binary (1 = urban residence)
- **Agricultural hukou**: Binary (1 = agricultural household registration)
- **Ln(income)**: Natural log of annual personal income
- **Survey year**: Fixed effects (factor)

S1.4 Missing Data

Missing values were imputed using the `missRanger` algorithm with predictive mean matching ($k = 5$). This method combines random forest imputation with nearest-neighbor matching to preserve the distributional properties of the original data. All variables in the analytic dataset were included in the imputation model.

S2. Heterogeneity Analysis: Full Estimates

Table S1 reports the full coefficient estimates for all Female × Moderator interactions presented graphically in the main text. Panel A contains individual-level moderators; Panel B contains provincial development indicators. All coefficients within each column come from a single OLS model that includes all interactions simultaneously. Joint F-tests for all nine interactions are significant in each domain ($p < .001$).

Table S1: Heterogeneity Analysis: Female × Moderator Interactions (HC1 Robust SEs)

Moderator × Female	Premarital Sex	Extramarital Sex	Same-Sex Relations
Panel A: Individual-level moderators			
Education (years)	−0.003	−0.002	0.013***
	(0.002)	(0.001)	(0.002)
Internet user	−0.119***	−0.040***	0.109***
	(0.020)	(0.012)	(0.016)
Married	0.045*	0.040**	−0.104***
	(0.019)	(0.013)	(0.017)
Religious	−0.036	−0.027	−0.006
	(0.028)	(0.017)	(0.022)
Urban residence	−0.028	−0.019	−0.028
	(0.021)	(0.013)	(0.017)
CCP member	0.113***	0.054**	0.105***
	(0.026)	(0.017)	(0.024)
Agricultural hukou	−0.005	0.027*	0.062***
	(0.021)	(0.013)	(0.018)
Panel B: Provincial development moderators			
Log(GDP per capita)	0.0899**	−0.0219	0.0764**
	(0.0329)	(0.0209)	(0.0276)
Provincial college rate (%)	−0.0033†	0.0006	−0.0008
	(0.0017)	(0.0011)	(0.0015)
Joint F-test	$F(9, 70{,}563) = 11.31^{***}$	$F(9, 70{,}563) = 8.37^{***}$	$F(9, 70{,}563) = 30.98^{***}$
Controls	Gender ideology (z), age, age^2/100, ln(income), wave FE		
N	70,593		

Note. Each column reports coefficients from a single OLS model that includes all Female × Moderator interactions simultaneously. Panel A: individual-level moderators. Panel B: provincial development indicators (GDP per capita from China Statistical Yearbook 2017; college rate from 2020 Population Census). HC1 robust standard errors in parentheses.
$^{\dagger}p < .10$; $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$.

S3. Cohort Heterogeneity: Full Estimates

Table S2 reports the Female × Cohort interaction coefficients presented graphically in the main text.

Table S2: Cohort × Gender Interactions (HC1 Robust SEs)

Cohort × Female	Premarital Sex (ref: 1920–1949)	Extramarital Sex	Same-Sex Relations
1950–1959	−0.034	−0.018	0.068***
	(0.024)	(0.014)	(0.019)
1960–1969	−0.053*	−0.035*	0.080***
	(0.024)	(0.015)	(0.018)
1970–1979	−0.122***	−0.081***	0.099***
	(0.025)	(0.016)	(0.020)
1980–1989	−0.189***	−0.078***	0.209***
	(0.027)	(0.017)	(0.023)
Post-1990	−0.157***	−0.069***	0.456***
	(0.029)	(0.019)	(0.028)
Joint F-test	$F(5, 78{,}076) = 15.70^{***}$	$F(5, 78{,}076) = 6.10^{***}$	$F(5, 78{,}076) = 52.84^{***}$
Controls		Full set + wave FE	
N		78,206	

Note. Each cell reports the coefficient on the Female × Cohort interaction term from a separate OLS regression. Reference cohort is 1920–1949. HC1 robust standard errors in parentheses. All models include the full control set, main effects of cohort dummies, and wave fixed effects.
$^{\dagger}p < .10$; $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$.

S4. Additional Figures

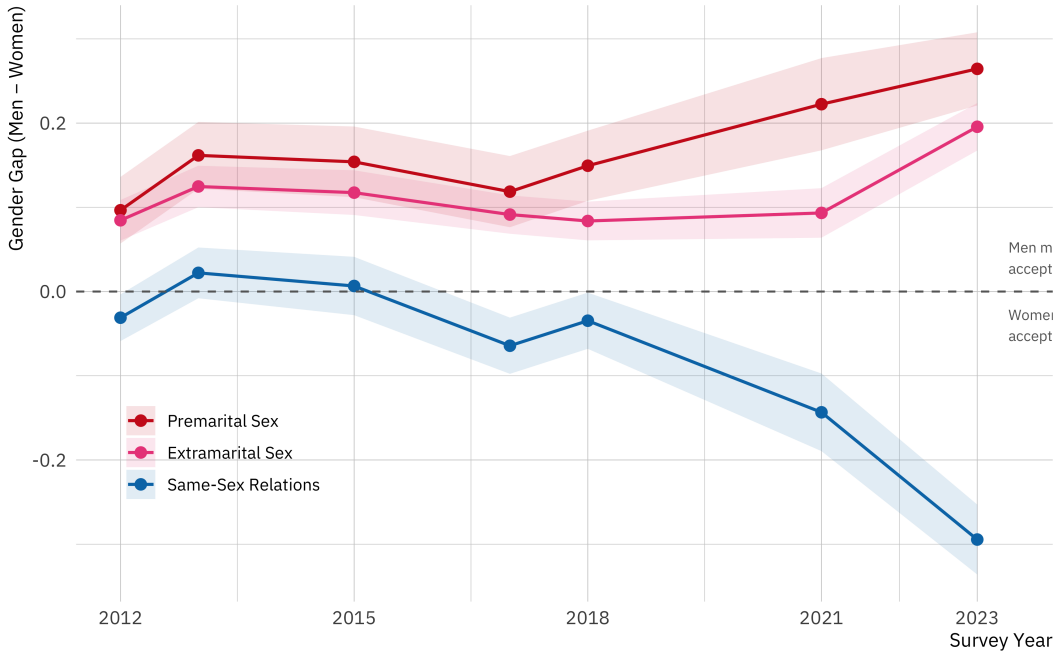

Figure S1: Gender gap (men − women) in sexual attitudes by survey year. Positive values indicate men are more accepting; negative values indicate women are more accepting. Shaded bands show 95% confidence intervals. The same-sex relations gap diverges sharply after 2017.

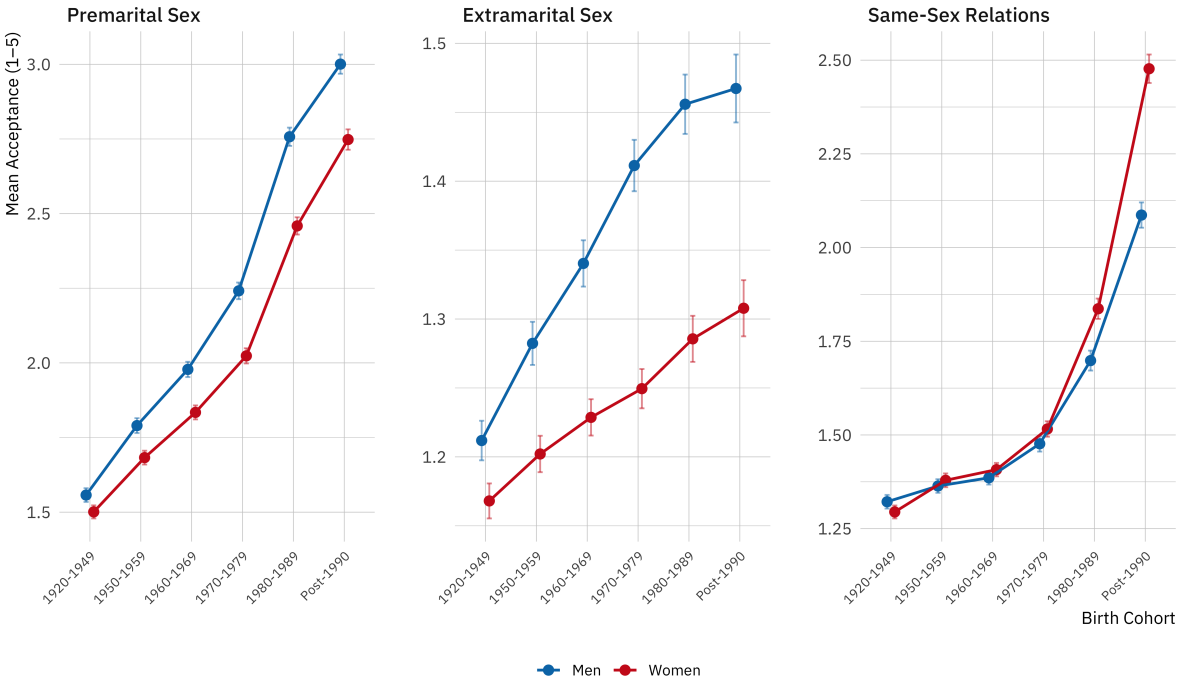

Figure S2: Mean sexual attitude acceptance by birth cohort and gender. Error bars show 95% confidence intervals. The gender gap in same-sex acceptance widens dramatically for post-1990 cohorts, driven primarily by women's increasing acceptance.

S5. Robustness and Sensitivity Analyses

This section consolidates all robustness checks and sensitivity analyses. We assess the stability of the main findings across alternative model specifications (ordered logistic regression, province fixed effects), alternative estimation strategies (propensity score matching, inverse probability weighting, listwise deletion), supplementary tests (gender ideology interaction, wave-specific estimates), and formal sensitivity analyses (causal mediation sensitivity, E-values).

S5.1 Ordered Logistic Regression

To verify that results are robust to the ordinal nature of the dependent variables, we re-estimated all main models using ordered logistic regression (proportional odds model).

Table S3: Ordered Logit Models: Robustness Check

	Premarital Sex	Extramarital Sex	Same-Sex Relations
Female	-0.222^{***}	-0.363^{***}	0.221^{***}
	(0.014)	(0.019)	(0.017)
Gender ideology (z)	-0.037^{***}	0.128^{***}	-0.110^{***}
	(0.008)	(0.010)	(0.009)
Age	-0.048^{***}	0.009^{*}	-0.073^{***}
	(0.003)	(0.004)	(0.003)
Age^2/100	0.016^{***}	-0.017^{***}	0.050^{***}
	(0.003)	(0.004)	(0.003)
Education (years)	0.022^{***}	0.005	0.035^{***}
	(0.002)	(0.003)	(0.003)
Married	-0.164^{***}	-0.347^{***}	-0.329^{***}
	(0.018)	(0.024)	(0.021)
CCP member	0.083^{***}	0.092^{***}	-0.042
	(0.019)	(0.025)	(0.022)
Religious	-0.071^{**}	-0.038	-0.194^{***}
	(0.023)	(0.031)	(0.028)
Internet user	0.350^{***}	0.351^{***}	0.052^{*}
	(0.019)	(0.027)	(0.023)
Urban	-0.068^{***}	-0.084^{***}	-0.057^{***}
	(0.004)	(0.006)	(0.005)
Agricultural hukou	-0.163^{***}	-0.133^{***}	-0.127^{***}
	(0.016)	(0.022)	(0.019)
Ln(income)	0.014^{***}	0.010^{***}	0.003
	(0.002)	(0.003)	(0.002)
Year FE	Yes	Yes	Yes
Log-likelihood	-96334.9	-50550.1	-64889.0
N	78,206	78,206	78,206

Note. Ordered logistic regression (proportional odds model). Coefficients are log-odds. $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$. Standard errors in parentheses.

S5.2 Province Fixed Effects

To address concerns about unobserved provincial heterogeneity, we re-estimated main models with province fixed effects.

Table S4: Province Fixed Effects Models: Robustness Check

	Premarital Sex	Extramarital Sex	Same-Sex Relations
Female	-0.135^{***}	-0.098^{***}	0.091^{***}
	(0.008)	(0.005)	(0.007)
Gender ideology (z)	-0.009^{*}	0.037^{***}	-0.047^{***}
	(0.004)	(0.003)	(0.004)
Age	-0.037^{***}	0.001	-0.039^{***}
	(0.001)	(0.001)	(0.001)
$\text{Age}^2/100$	0.018^{***}	-0.003^{***}	0.028^{***}
	(0.001)	(0.001)	(0.001)
Education (years)	0.008^{***}	0.000	0.015^{***}
	(0.001)	(0.001)	(0.001)
Married	-0.102^{***}	-0.093^{***}	-0.153^{***}
	(0.010)	(0.006)	(0.008)
CCP member	0.039^{***}	0.030^{***}	-0.040^{***}
	(0.011)	(0.007)	(0.009)
Religious	0.000	-0.002	-0.055^{***}
	(0.013)	(0.008)	(0.011)
Internet user	0.164^{***}	0.081^{***}	-0.017
	(0.011)	(0.007)	(0.009)
Urban	-0.041^{***}	-0.021^{***}	-0.027^{***}
	(0.003)	(0.002)	(0.002)
Agricultural hukou	-0.070^{***}	-0.023^{***}	-0.041^{***}
	(0.009)	(0.006)	(0.008)
Ln(income)	0.005^{***}	0.001^{*}	-0.002
	(0.001)	(0.001)	(0.001)
Year FE	Yes	Yes	Yes
Province FE	Yes	Yes	Yes
R^2	0.189	0.040	0.133
N	78,206	78,206	78,206

Note. OLS with province fixed effects (dummy variables). Province coefficients not shown. $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$. Standard errors in parentheses.

S5.3 Gender Ideology × Gender Interaction

To formally test whether gender ideology operates differently for men and women, we estimated models with a female × gender ideology interaction term.

Table S5: Gender Ideology × Gender Interaction Models

	Premarital Sex	Extramarital Sex	Same-Sex Relations
Female	-0.130^{***}	-0.097^{***}	0.091^{***}
	(0.008)	(0.005)	(0.007)
Gender ideology (z)	-0.002	0.046^{***}	-0.000
	(0.006)	(0.004)	(0.005)
Female × Gender ideology (z)	-0.016^{*}	-0.015^{**}	-0.085^{***}
	(0.008)	(0.005)	(0.006)
Age	-0.036^{***}	0.001	-0.038^{***}
	(0.001)	(0.001)	(0.001)
$\text{Age}^2/100$	0.018^{***}	-0.003^{***}	0.028^{***}
	(0.001)	(0.001)	(0.001)
Education (years)	0.010^{***}	0.000	0.014^{***}
	(0.001)	(0.001)	(0.001)
Married	-0.111^{***}	-0.101^{***}	-0.159^{***}
	(0.010)	(0.006)	(0.008)
CCP member	0.041^{***}	0.030^{***}	-0.041^{***}
	(0.011)	(0.007)	(0.009)
Religious	-0.026^{*}	-0.002	-0.067^{***}
	(0.012)	(0.008)	(0.010)
Internet user	0.182^{***}	0.087^{***}	-0.012
	(0.011)	(0.007)	(0.009)
Urban	-0.046^{***}	-0.021^{***}	-0.030^{***}
	(0.002)	(0.002)	(0.002)
Agricultural hukou	-0.090^{***}	-0.033^{***}	-0.051^{***}
	(0.009)	(0.006)	(0.008)
Ln(income)	0.007^{***}	0.002^{***}	-0.001
	(0.001)	(0.001)	(0.001)
Year FE	Yes	Yes	Yes
R^2	0.176	0.031	0.127
N	78,206	78,206	78,206

Note. OLS regression testing whether the effect of gender ideology on sexual attitudes differs by gender. The interaction term (Female × Gender ideology) captures the differential effect. $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$. Standard errors in parentheses.

S5.4 Wave-Specific Estimates

To assess the stability of key coefficients across survey waves, we estimated the main model specification separately for each CGSS wave (2012–2023). Figure S3 plots the female coefficient and gender ideology coefficient by wave.

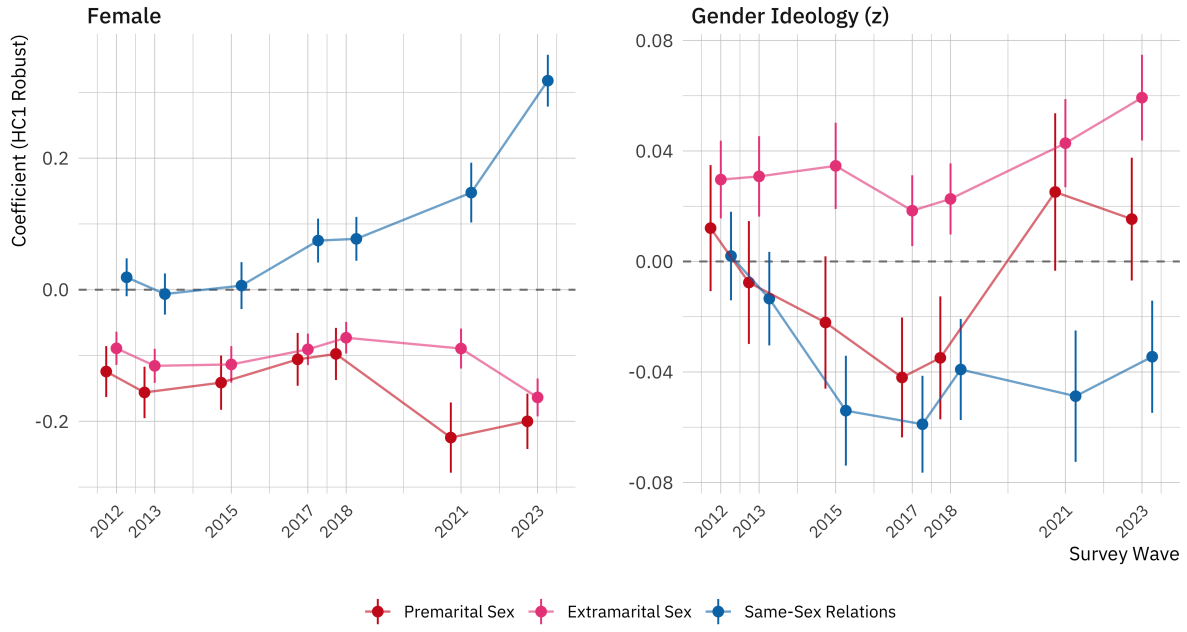

Figure S3: Wave-specific estimates of the Female coefficient and Gender Ideology coefficient. Each point represents the coefficient from a model estimated on a single survey wave. Error bars show 95% confidence intervals.

S5.5 Causal Mediation Sensitivity Analysis

Following Imai et al. (2010), we conducted sensitivity analysis for the causal mediation estimates using the `medsens()` function from the `mediation` R package. The sensitivity parameter $\tilde{\rho}$ represents the correlation between the residuals of the mediator and outcome models. We report the value of $\tilde{\rho}$ at which the ACME equals zero: $\tilde{\rho} = -0.010$ for premarital sex, $\tilde{\rho} = 0.050$ for extramarital sex, and $\tilde{\rho} = -0.050$ for same-sex relations. These values are small, indicating that the mediation estimates are sensitive to unmeasured confounding: a confounder inducing a residual correlation of approximately 0.05 would suffice to nullify the extramarital and same-sex mediation effects.

S5.6 E-Value Sensitivity Analysis

Following VanderWeele and Ding (2017), we calculated E-values for the female coefficients in each domain. The E-value for the premarital sex coefficient is 1.48 (CI limit: 1.45), for extramarital sex 1.56 (CI limit: 1.52), and for same-sex relations 1.39 (CI limit: 1.35). These moderate thresholds suggest that unmeasured confounding alone is unlikely to fully explain the observed gender gaps, though they indicate that the same-sex domain is somewhat more vulnerable to confounding explanations.

S5.7 Summary of All Robustness Specifications

Table S7 summarizes the female coefficient across all robustness specifications. The direction, magnitude, and statistical significance of the gender gap in each domain are consistent across ordered logistic regression, province fixed effects, propensity score matching (1:1 nearest-neighbor with caliper 0.2, $N = 65{,}350$), inverse probability weighting (stabilized, trimmed at 99th percentile), and listwise deletion without imputation. E-values and sensitivity parameters from mediation analysis are also reported. We note that because gender is not a manipulable treatment, the PSM and IPW analyses should be interpreted as sensitivity checks for covariate

imbalance rather than as causal treatment effect estimates.

Table S6: Summary of All Robustness Specifications

Specification	Premarital Sex	Extramarital Sex	Same-Sex Relations	N
Main results				
OLS with HC1 SE	-0.145***	-0.103***	0.085***	78,206
	(0.008)	(0.005)	(0.007)	
Ordered logistic regression	-0.249***	-0.386***	0.203***	78,206
	(0.014)	(0.019)	(0.017)	
Addressing geographic confounding				
Province fixed effects	-0.148***	-0.104***	0.085***	78,206
	(0.008)	(0.005)	(0.007)	
Addressing selection on observables				
Propensity score matching (1:1)	-0.149***	-0.105***	0.084***	65,350
	(0.008)	(0.005)	(0.007)	
Inverse probability weighting	-0.141***	-0.104***	0.086***	78,206
	(0.008)	(0.005)	(0.007)	
Sample restrictions				
Listwise deletion (no imputation)	-0.145***	-0.103***	0.085***	78,206
	(0.008)	(0.005)	(0.007)	
Sensitivity measures				
E-value (point estimate)	1.48	1.56	1.39	
E-value (CI limit)	1.45	1.52	1.35	
Medsens $\tilde{\rho}$ at ACME = 0	-0.010	0.051	-0.047	

Note. All models include controls for gender ideology (std.), age, age^2/100, education, marital status, CCP membership, religion, Internet use, urban residence, agricultural *hukou*, log income, and wave fixed effects. HC1 robust standard errors in parentheses (except PSM, which uses matching weights). The female coefficient is statistically significant across all specifications.
$^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$.

References

Imai, K., Keele, L., and Yamamoto, T. (2010). Identification, inference and sensitivity analysis for causal mediation effects. *Statistical Science*, 25(1):51–71.

VanderWeele, T. J. and Ding, P. (2017). Sensitivity analysis in observational research: Introducing the E-value. *Annals of Internal Medicine*, 167(4):268–274.